

Projects Portfolio

by Ahmad Fauzan Rinaufaldi

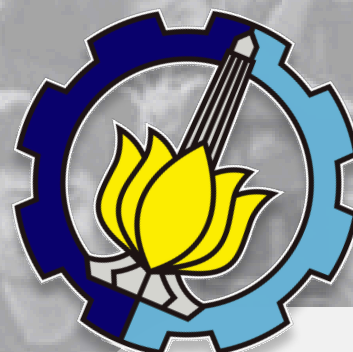
About

**Information Systems Graduate
Specializing in Data Analytics & Data Engineering
Institut Teknologi Sepuluh Nopember**

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Ahmad Fauzan Rinaufaldi



Education :

- Institut Teknologi Sepuluh Nopember (ITS): Aug 2022 – Sep 2026 (Expected)
- SMA Muhammadiyah 2 Surabaya: Jul 2019 – Jun 2022



Working Experience

- PT. Telekomunikasi Selular (Telkomsel): Jul 2025 – Dec 2025
- PT. Pelindo Marine Service : Jan 2025 – Feb 2025

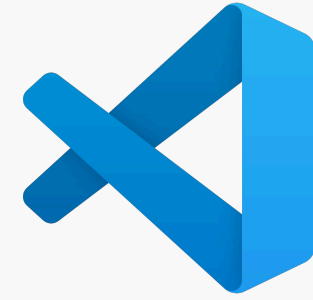
Organizational Experience :

- Information Systems Expo (ISE!) 2024 – Project Officer: Jan 2024 – Dec 2024
- Society of Renewable Energy (SRE) ITS SC – Competency Development Staff: Sep 2023 – Jun 2024
- ISC Dies Natalis ITS 63 – Event Staff: Aug 2023 – Oct 2023
- HMSI ITS – Intern Staff of Internal Affairs: Jul 2023 – Oct 2024
- Information Systems Expo (ISE!) 2023 – Paradigm Event FOH Staff: Mar 2023 – Oct 2023
- GERIGI ITS 2023 – Main Event Staff: Jul 2023 – Aug 2023
- TEDxITS – Logistics and Consumption Staff : Feb 2023 – May 2023
- Future Energy Summit (FUTUREST) 2023 : Nov 2022 – May 2023
- Ini Lho ITS! – Logistics, Tools, and Equipment Staff: Oct 2022 – Jan 2023

Software Usage



Python



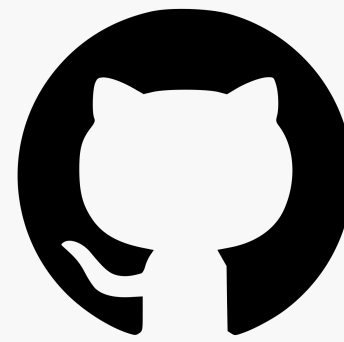
Visual Studio Code



Power BI



Figma



GitHub



Excel



Google Colab



DBeaver



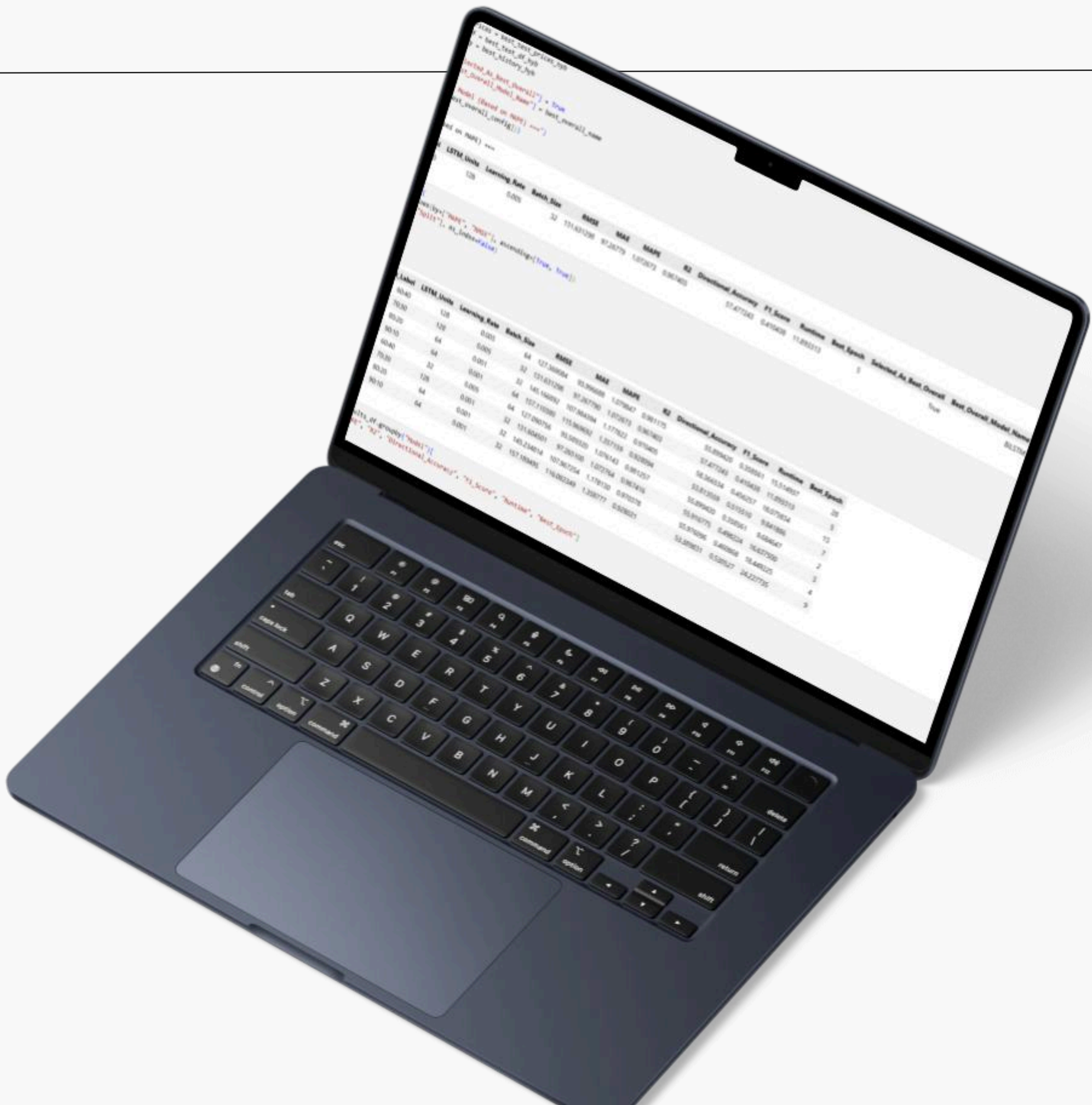
Protege



Pentaho DI



Final Thesis: Stock Price **Forecasting** Using Hybrid Method **Bidirectional Long Short Term Memory** and **Multi Head Attention** Study Case **BBCA.JK**





Time Series Forecasting, Deep Learning, Financial Data Analysis

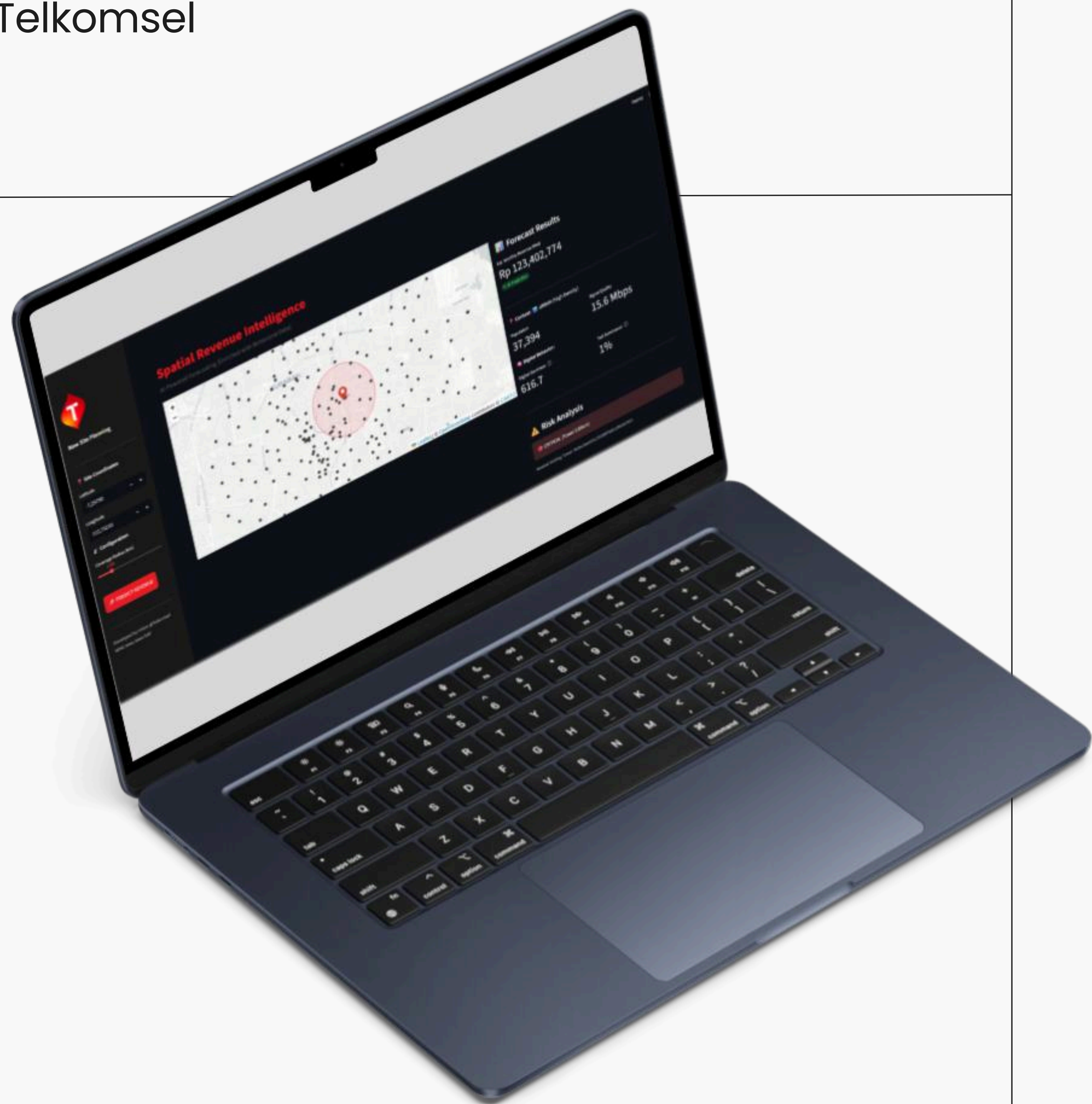
Developed a deep learning-based stock forecasting model using a hybrid Bidirectional Long Short Term Memory (BiLSTM) with Multi Head Attention Mechanism to predict BBKA stock price movements.

Key tasks included constructing a multivariate time series dataset, transforming price data into return for improved learning stability, and optimizing model performance through hyperparameter tuning (LSTM units, learning rate, batch size, and data split).

The final model demonstrates strong predictive accuracy and generalization capability, supported by evaluation metrics and visualization to aid investment decision-making.

Access the documents here :
intip.in/FinalThesis

Telkomsel Internship Project : Spatial **Revenue Forecasting** Smart **BTS** Simulation for Telkomsel





Geospatial Forecasting, Machine Learning,
Industrial Data Analysis

Built a machine learning model using PyCaret and LightGBM to predict the monthly revenue of new cell towers.

Key tasks included mapping out 779,399 grid tiles to define specific service areas, and training the model using practical data like population density, network speed, and competitor strength.

The final model achieved a high accuracy rate with just a 1.861% error margin (MAPE). It is packaged into a user-friendly Streamlit dashboard that simulates coverage areas and calculates financial risks to guide better investment planning.

Route Optimization for DKI Jakarta City Park Cleaning Inspections by the Environmental Agency using Soft Computing Methods (GA, ACO, ACO+2-Opt) on the Vehicle Routing Problem



Urban Computing, Route Optimization, Data Visualization

Developed as an individual project for a Soft Computing course, this system models city park inspections in Jakarta using Genetic Algorithm, Ant Colony Optimization, and a hybrid ACO with 2-opt local search, built around open-source park data and a self-constructed scenario, solved as a VRPTW.

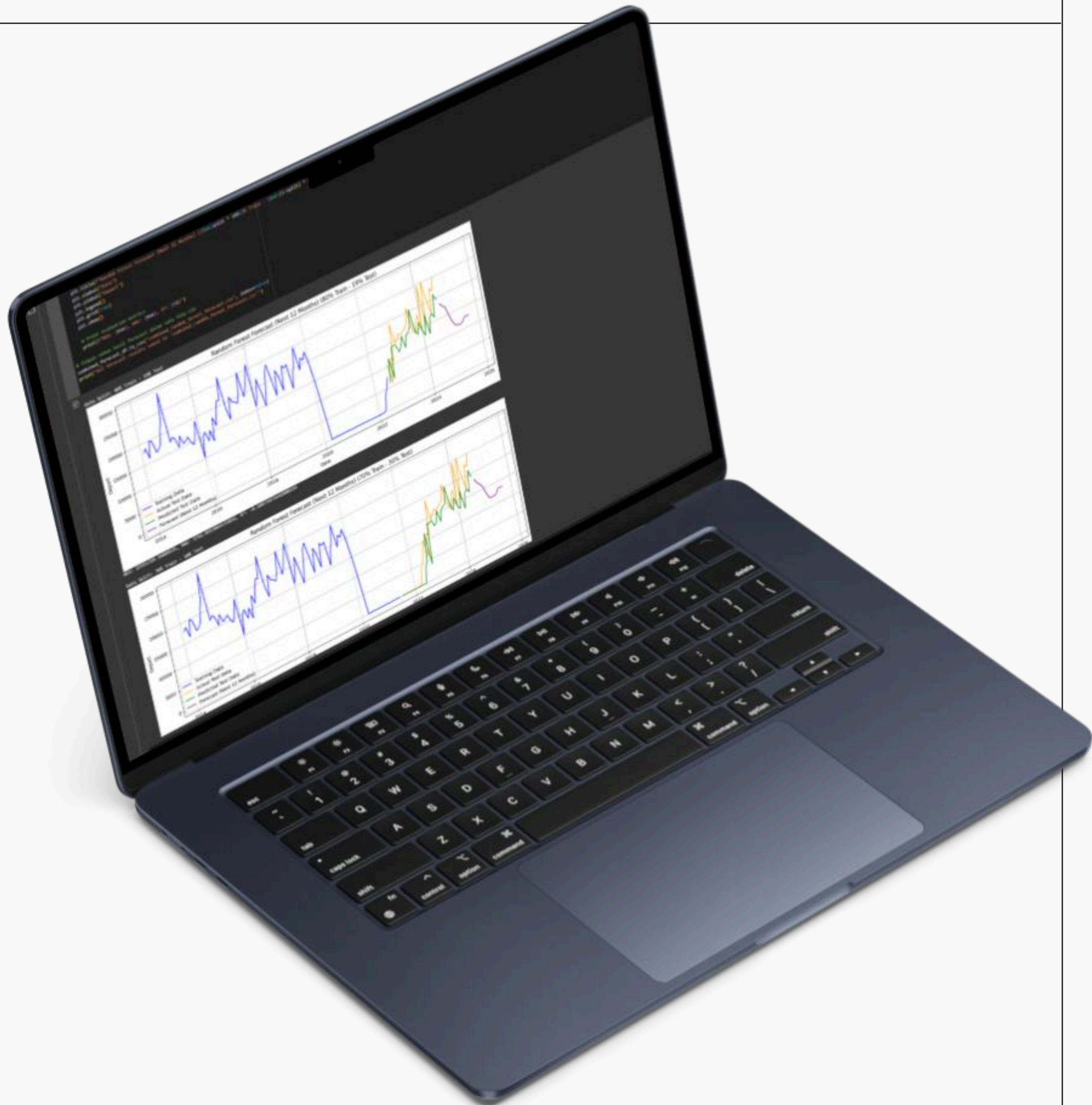
The scenario applies realistic constraints: an 8-hour operational window, vehicle speeds of 30–50 km/h based on traffic density, and 30-minute inspection durations. K-Means clustering zoned the city first to improve scalability.

The resulting system delivers route visualizations, convergence analysis, and resource evaluation, showing how soft computing methods can support more efficient inspection planning.

Access the documents here :

intip.in/RouteOptimization

Time Series Forecasting: Predicting Domestic Passenger Arrivals at Kualanamu International Airport, North Sumatra





Forecasting, Data Cleansing, Data Visualization

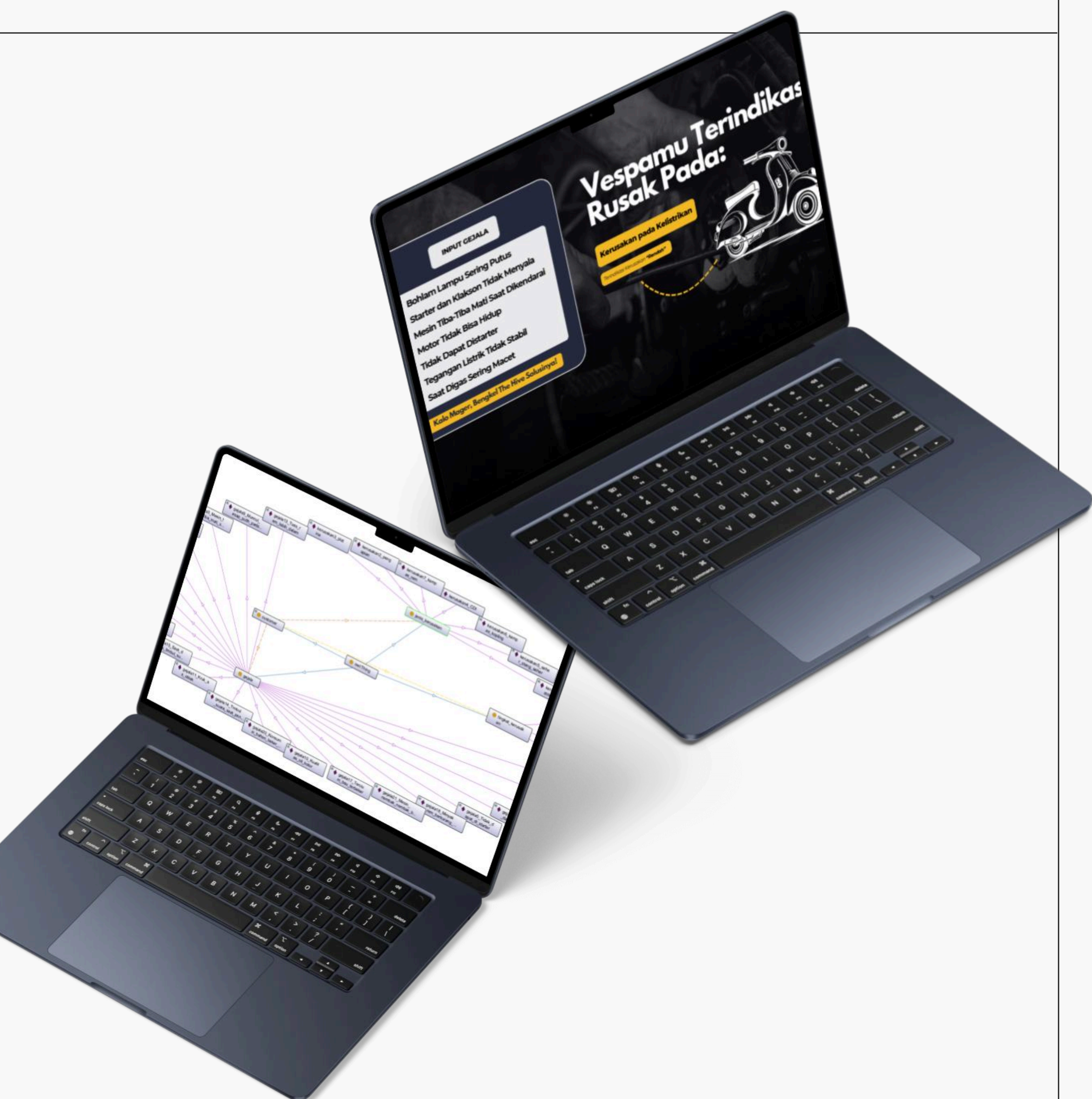
Developed as a team project, this time series forecasting system predicts monthly domestic passenger arrivals at Kualanamu International Airport using SARIMA, Random Forest, and a hybrid GRU-Random Forest model.

The project evaluated multiple train-test scenarios (80:20, 70:30, 60:40) using MSE, MAE, MAPE, and R^2 metrics. Each model was benchmarked under different data splits to thoroughly test its robustness under limited or imbalanced data.

The resulting system demonstrates that Random Forest achieved the highest predictive accuracy, while SARIMA excelled at capturing seasonal patterns. This solution effectively supports strategic airport planning through data-driven demand forecasting.

Access the documents here :
intip.in/TimeSeriesForecasting

Application of **Ontology Based Expert System** for Diagnosis of Vespa Scooter Using **Protégé**





Ontology Modeling, Knowledge Representation,
SWRL Rule, UI Design

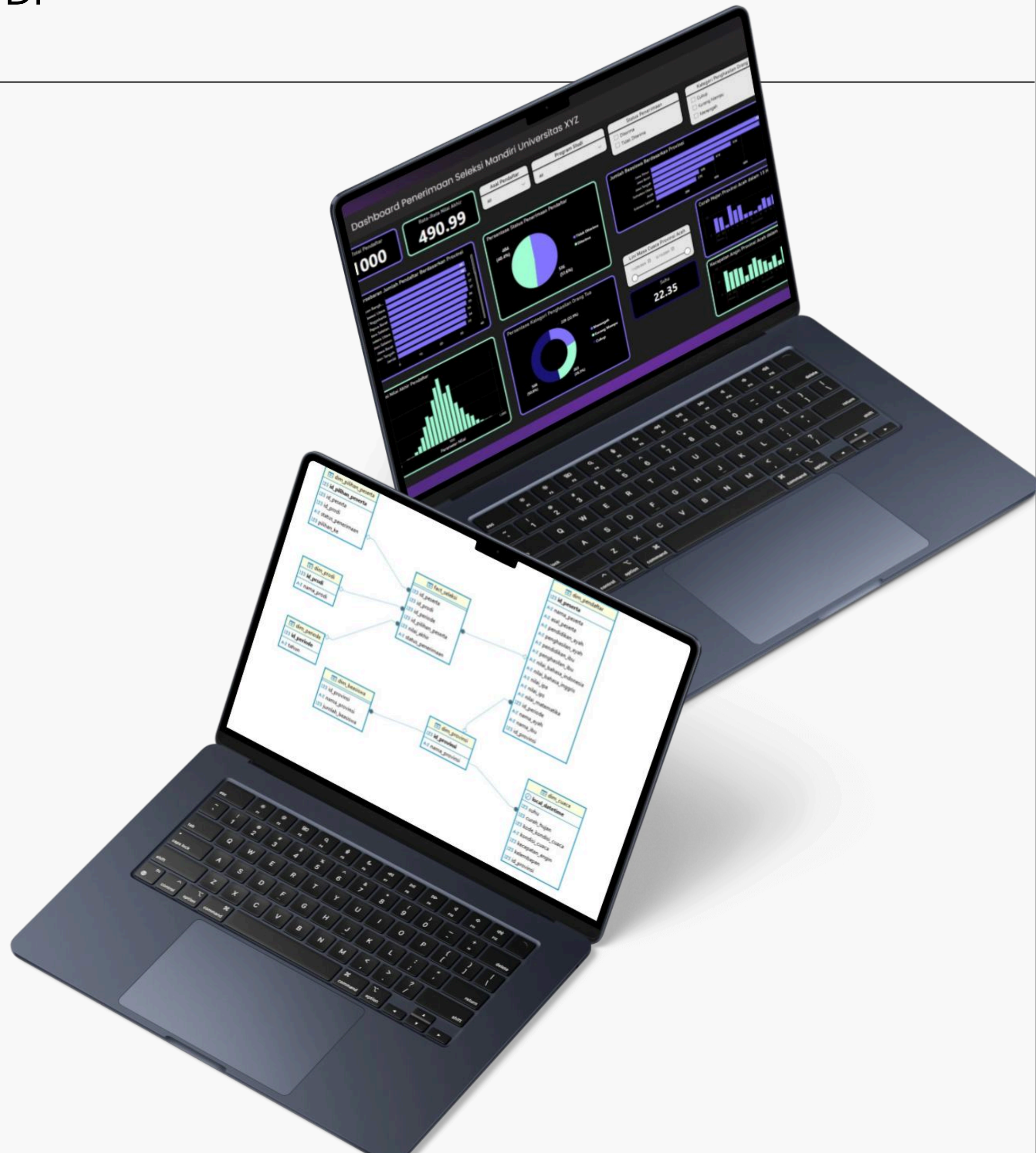
Developed as a team project, this ontology-driven expert system diagnoses Vespa scooter faults, built with mechanics at a real service center and modeled in Protégé to map symptoms, components, and fault types.

SWRL rules enable logical inference from observed symptoms, letting the system recommend fault categories and suggest preliminary repair actions grounded in real practitioner knowledge. A supporting UI was prototyped in Figma to visualize the diagnostic workflow.

The resulting system shows how ontology-based reasoning can encode real mechanical expertise into a structured, explainable diagnostic tool.

Access the documents here :
[**intip.in/Ontology**](https://intip.in/Ontology)

Microsoft **Power BI Dashboard** using **Structured** and **Unstructured Data** from DBeaver and Pentaho DI



SQL Query, ETL, Data Visualization, Data Integration

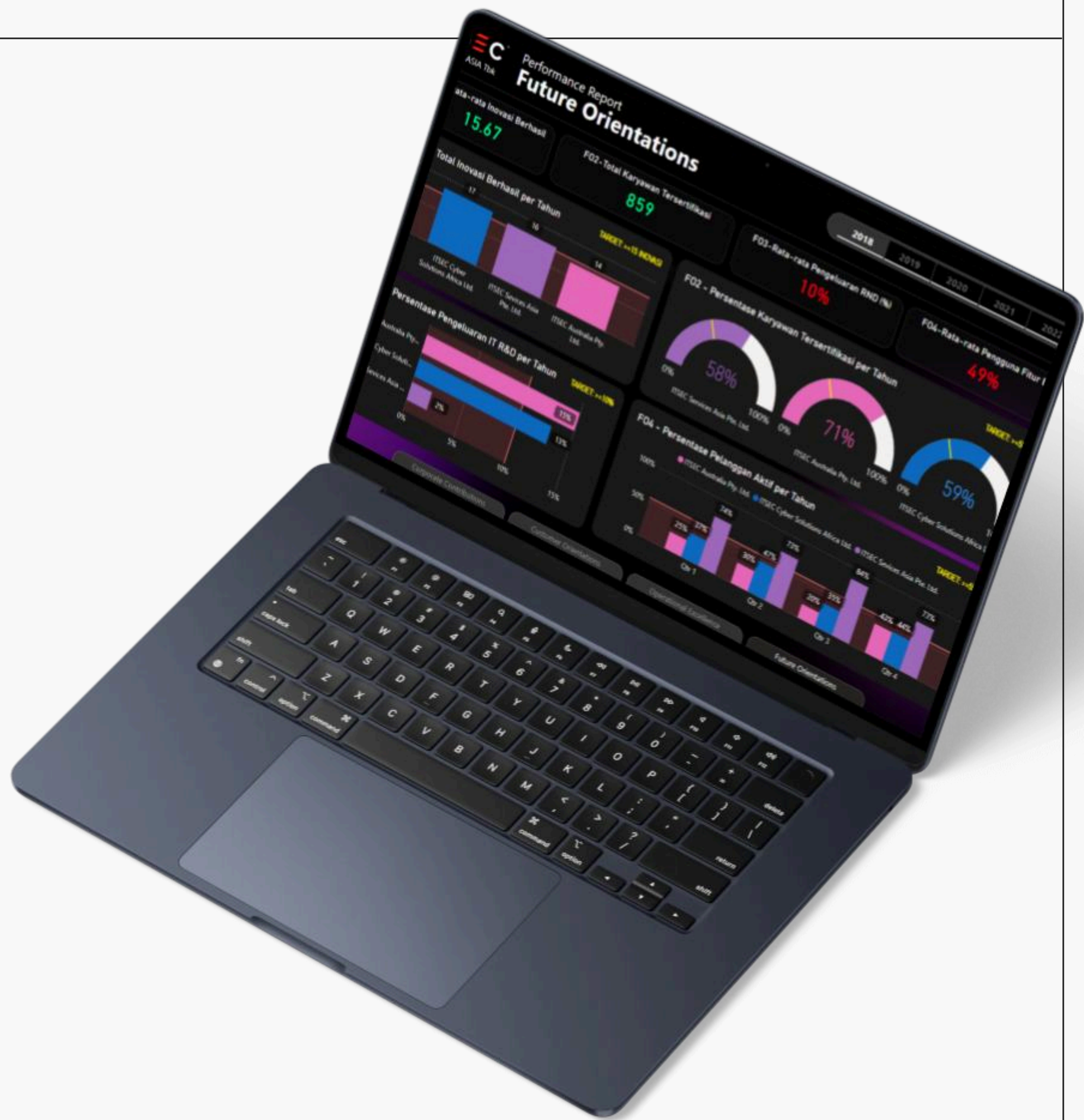
Developed as an individual project, this interactive Power BI dashboard integrates scholarship admissions data, general university admissions data, and BMKG weather forecasts for Aceh to support data-driven scholarship allocation decisions.

I designed a star schema data warehouse in DBeaver, structuring fact and dimension tables around the admissions process, and handled transformations via Pentaho Data Integration (PDI). The dashboard visualizes applicant and scholarship distribution by region, income category, and score, with weather layered in as context.

The resulting dashboard enables interactive cross-filtering, giving scholarship organizations a tool to assess eligibility, trends, and regional distribution for more effective aid allocation.

Access the documents here :
intip.in/DataLakehouse

Performance Monitoring **Dashboard Development** Using **IT Balanced Scorecard** and **Power BI** at PT ITSEC





Business Intelligence, Power BI Dashboarding, Data Modelling,
Data Visualization, Strategic and hOperation Analysis

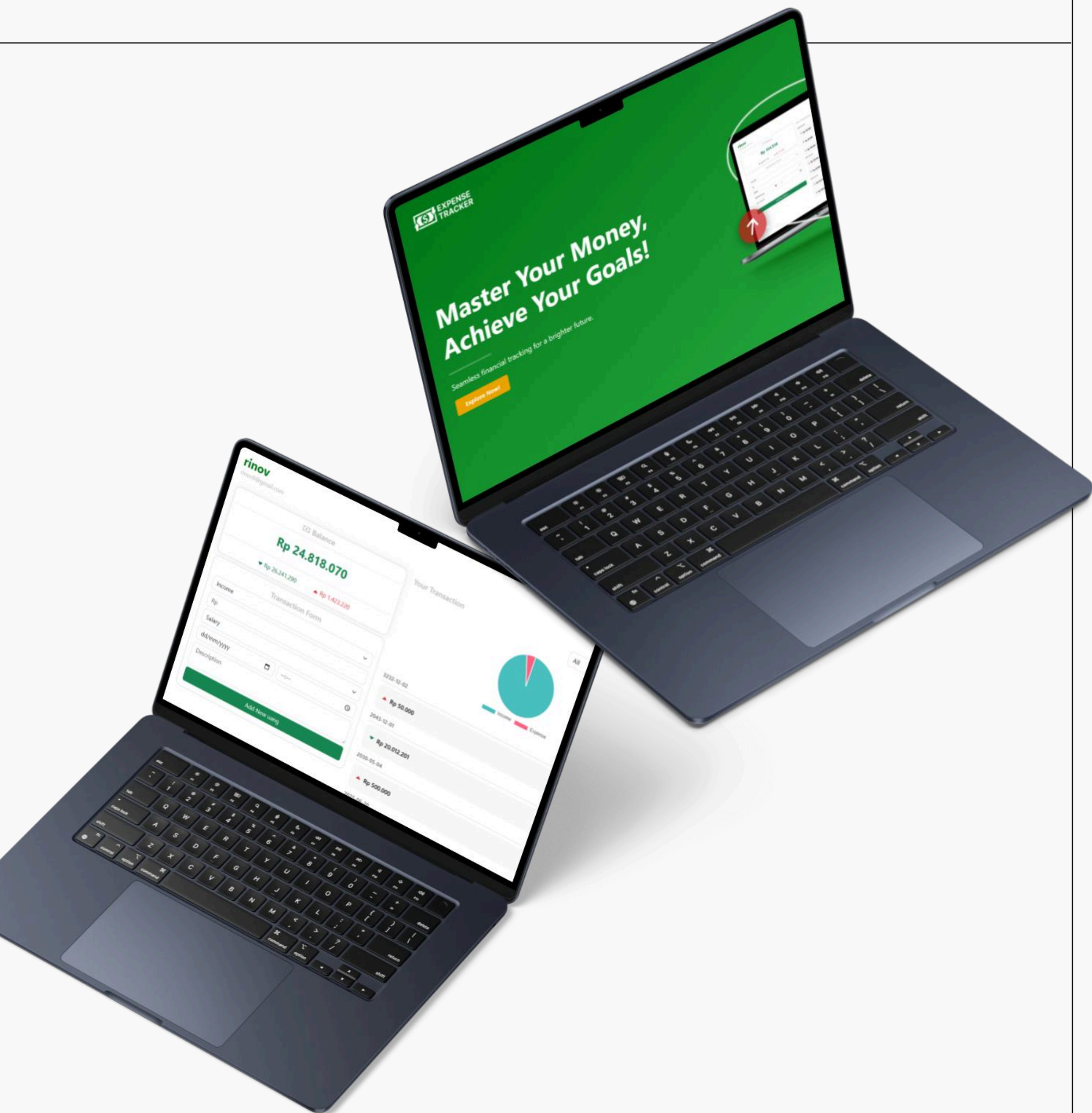
Developed as a team project, this interactive Power BI dashboard applies the IT Balanced Scorecard (IT-BSC) framework to evaluate business performance at PT ITSEC Asia, covering Jakarta HQ and two branch offices.

The project involved simulating 16 strategic KPIs across four IT-BSC perspectives: Corporate Contribution, Customer Orientation, Operational Excellence, and Future Orientation. Metrics were generated via Mockaroo, structured in MySQL and DBeaver, and integrated into Power BI.

The resulting dashboard delivers localized branch views to analyze service availability, cybersecurity incidents, client satisfaction, and project delivery—empowering management with data-driven oversight for technology-driven operations.

Access the documents here :
intip.in/ITSECDashboard

FullstackExpense: Modern Expense Tracker with End-to-End **DevOps** using CI/CD & Firebase Backend





Fullstack WebDev, DevOps Engineering, UI/UX Design, Cloud Deployment, API Integration

Developed as a team project, this fullstack expense tracker optimizes personal finance management through backend migration to Firebase and integration of a complete DevOps pipeline.

The project involved refining the frontend with dynamic input formatting and real-time balance updates, while configuring a CI/CD pipeline via GitHub Actions for linting, testing, and Docker containerization. The application utilizes Firebase for authentication and Firestore, with cloud hosting deployed on Google Cloud Run.

The resulting system demonstrates a robust end-to-end DevOps workflow, combining frontend-backend integration with automated testing pipelines to deliver a reliable and streamlined UI.

Access the documents here :
intip.in/DevopsProject

e-CCWS: A **CCWS Booking App** to Save the Hustle of CCWS





User Research, Needfinding, UI/UX Design, User Flow, WireFrame, Visual Storytelling

Developed as a team project, this full redesign of the Mamikos application optimizes the property-hunting experience based on user-centered research and needfinding interviews with tenants and owners.

The project involved prototyping from low-fidelity to high-fidelity in Figma, implementing key solutions such as real-time filtering, information renewal status, and media-sharing chat features. Evaluations were conducted using heuristic analysis, user testing, and "Wizard of Oz" simulations.

The resulting prototype supports clearer property listings and more intuitive search experiences, effectively empowering diverse user personas from students to property managers.

Access the documents here :
intip.in/RedesignMamikos

Wildfire, Weather, and Rainfall Analytics in Western Australia: **Data Clustering** and **Visualization**





Data Collection, Clustering Analysis,
Data Visualization

Developed as a team project, this diagnostic analytics project evaluates wildfire frequency, temperature variation, and risk clustering across multiple vulnerable suburbs in Western Australia.

The project integrated official weather datasets from the Bureau of Meteorology (BOM) with historical incident data, applying clustering techniques to classify suburb-level fire risks. Regression analysis was also conducted to analyze the correlation between maximum temperatures and fire frequency across different seasons.

The resulting analysis highlights critical heatwave patterns and bushfire-prone zones, delivering evidence-based insights to support disaster response planning and public awareness in high-risk regions.

Access the documents here :
intip.in/BushfireAnalysisData

Thank- you.

SISTEM INFORMASI
DEPARTMENT OF INFORMATION SYSTEMS

Ahmad Fauzan Rinaufaldi ↗

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